

Strict asymptotic nilpotency in cellular automata

Ville Salo¹

¹University of Turku, Finland

Automata 2017

Outline

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

The questions

The questions

Experimentation

Experimentation

1D

1D

d D

d D

Groups and graphs

Groups and graphs

Conclusions

Conclusions

Some questions

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

Question

Which one-dimensional sofic shifts support cellular automata which are asymptotically nilpotent but not nilpotent?

The questions

Experimentation

1D

dD

Groups and graphs

Conclusions

Question

Which multidimensional SFTs support cellular automata that are asymptotically nilpotent but not nilpotent? Do any?

Question

Which full shifts on finitely generated groups support cellular automata that are asymptotically nilpotent but not nilpotent? Do any?

A nilpotent CA on the binary full shift

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

- Define $f : \{0, 1\}^{\mathbb{Z}} \rightarrow \{0, 1\}^{\mathbb{Z}}$ by

$$\forall i \in \mathbb{Z} : f(x)_i = 1 \iff x_{[i-2, i+2]} \in \{00111, 11100\}$$

for all $x \in \{0, 1\}^{\mathbb{Z}}$.

The questions

Experimentation

1D

dD

Groups and graphs

Conclusions

A nilpotent CA on the binary full shift

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

The questions

Experimentation

1D

dD

Groups and graphs

Conclusions

- Define $f : \{0, 1\}^{\mathbb{Z}} \rightarrow \{0, 1\}^{\mathbb{Z}}$ by

$$\forall i \in \mathbb{Z} : f(x)_i = 1 \iff x_{[i-2, i+2]} \in \{00111, 11100\}$$

for all $x \in \{0, 1\}^{\mathbb{Z}}$.

- This is a cellular automaton (on the binary full shift $\{0, 1\}^{\mathbb{Z}}$).

A nilpotent CA on the binary full shift

- Define $f : \{0, 1\}^{\mathbb{Z}} \rightarrow \{0, 1\}^{\mathbb{Z}}$ by

$$\forall i \in \mathbb{Z} : f(x)_i = 1 \iff x_{[i-2, i+2]} \in \{00111, 11100\}$$

for all $x \in \{0, 1\}^{\mathbb{Z}}$.

- This is a cellular automaton (on the binary full shift $\{0, 1\}^{\mathbb{Z}}$).
- Here's a spacetime diagram: (0 = white, 1 = black, time goes down, first row x , second $f(x)$, third $f^2(x)$)



A nilpotent CA on the binary full shift

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

The questions

Experimentation

1D

dD

Groups and graphs

Conclusions

- Define $f : \{0, 1\}^{\mathbb{Z}} \rightarrow \{0, 1\}^{\mathbb{Z}}$ by

$$\forall i \in \mathbb{Z} : f(x)_i = 1 \iff x_{[i-2, i+2]} \in \{00111, 11100\}$$

for all $x \in \{0, 1\}^{\mathbb{Z}}$.

- This is a cellular automaton (on the binary full shift $\{0, 1\}^{\mathbb{Z}}$).
- Here's a spacetime diagram: (0 = white, 1 = black, time goes down, first row x , second $f(x)$, third $f^2(x)$)



- This CA is *nilpotent*, that is,
 $\exists n \in \mathbb{N} : \forall x \in \{0, 1\}^{\mathbb{Z}} : f^n(x) = 0^{\mathbb{Z}}$.

Experiment 1: A random CA

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

The questions

Experimentation

1D

dD

Groups and graphs

Conclusions

- Define $f_1 : \{0, 1\}^{\mathbb{Z}} \rightarrow \{0, 1\}^{\mathbb{Z}}$ by
 $f_1(x)_i = 1 \iff x_{[i-3, i+3]} \in K_1$, where

$K_1 = \{0010000, 1010000, 0001000, 1001000, 0011000, 1000100, 0010100, 0110100, 1101100, 1011100,$
 $0111100, 0000010, 0100010, 1100010, 0010010, 0011010, 0000110, 0101110, 0011110, 1011110,$
 $1100001, 0010001, 0110001, 0001001, 1001001, 1011001, 1111001, 0000101, 0010101, 0110101,$
 $1101101, 0011101, 1111101, 1000011, 0010011, 0110011, 0001011, 1001011, 0011011, 0100111, 0010111\}$

was picked at random.

Experiment 1: A random CA

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

The questions

Experimentation

1D

dD

Groups and graphs

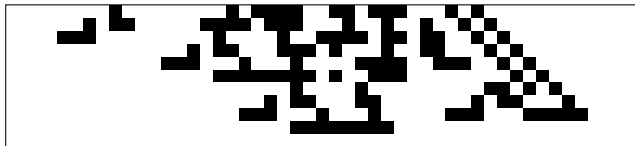
Conclusions

- Define $f_1 : \{0, 1\}^{\mathbb{Z}} \rightarrow \{0, 1\}^{\mathbb{Z}}$ by
 $f_1(x)_i = 1 \iff x_{[i-3, i+3]} \in K_1$, where

$K_1 = \{0010000, 1010000, 0001000, 1001000, 0011000, 1000100, 0010100, 0110100, 1101100, 1011100, 0111100, 0000010, 0100010, 1100010, 0010010, 0011010, 0000110, 0101110, 0011110, 1011110, 1100001, 0010001, 0110001, 0001001, 1001001, 1011001, 1111001, 0000101, 0010101, 0110101, 1101101, 0011101, 1111101, 1000011, 0010011, 0110011, 0001011, 1001011, 0011011, 0100111, 0010111\}$

was picked at random.

- Here's a spacetime diagram:



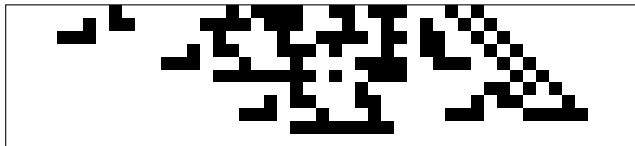
Experiment 1: A random CA

- Define $f_1 : \{0, 1\}^{\mathbb{Z}} \rightarrow \{0, 1\}^{\mathbb{Z}}$ by
 $f_1(x)_i = 1 \iff x_{[i-3, i+3]} \in K_1$, where

$K_1 = \{0010000, 1010000, 0001000, 1001000, 0011000, 1000100, 0010100, 0110100, 1101100, 1011100,$
 $0111100, 0000010, 0100010, 1100010, 0010010, 0011010, 0000110, 0101110, 0011110, 1011110,$
 $1100001, 0010001, 0110001, 0001001, 1001001, 1011001, 1111001, 0000101, 0010101, 0110101,$
 $1101101, 0011101, 1111101, 1000011, 0010011, 0110011, 0001011, 1001011, 0011011, 0100111, 0010111\}$

was picked at random.

- Here's a spacetime diagram:



- This particular configuration died after finitely many steps. Do all?

Strict asymptotic nilpotency in cellular automata

A perhaps nilpotent CA on the binary full shift

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

The questions

Experimentation

1D

dD

Groups and graphs

Conclusions

- Define $f_2 : \{0, 1\}^{\mathbb{Z}} \rightarrow \{0, 1\}^{\mathbb{Z}}$ by
 $f_2(x)_i = 1 \iff x_{[i-3, i+3]} \in K_2$, where

$K_2 = \{0010000, 1010000, 1110000, 0001000, 1010100, 1001100, 1101100, 1011100, 0000010, 1000010, 1100010, 0010010, 0101010, 1101010, 0111010, 1010110, 0101110, 1011110, 0111110, 1111110, 0100001, 1010001, 0110001, 1110001, 0001001, 0011001, 1011001, 0000101, 1000101, 1010101, 0110101, 1101101, 1011101, 0111101, 1000011, 0100011, 1001011, 1101011, 1011011, 0111011, 0000111, 0100111, 0110111, 0011111\}$



A perhaps nilpotent CA on the binary full shift

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

The questions

Experimentation

1D

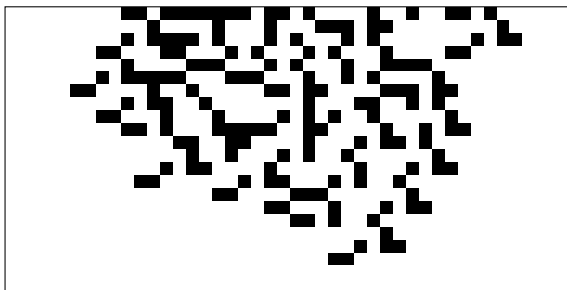
dD

Groups and graphs

Conclusions

- Define $f_2 : \{0, 1\}^{\mathbb{Z}} \rightarrow \{0, 1\}^{\mathbb{Z}}$ by
 $f_2(x)_i = 1 \iff x_{[i-3, i+3]} \in K_2$, where

$K_2 = \{0010000, 1010000, 1110000, 0001000, 1010100, 1001100, 1101100, 1011100, 0000010, 1000010, 1100010, 0010010, 0101010, 1101010, 0111010, 1010110, 0101110, 1011110, 0111110, 1111110, 0100001, 1010001, 0110001, 1110001, 0001001, 0011001, 1011001, 0000101, 1000101, 1010101, 0110101, 1101101, 1011101, 0111101, 1000011, 0100011, 1001011, 1101011, 1011011, 0111011, 0000111, 0100111, 0110111, 0011111\}$



- So is this one nilpotent?

Nope: f_2 has eventually periodic points!

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

The questions

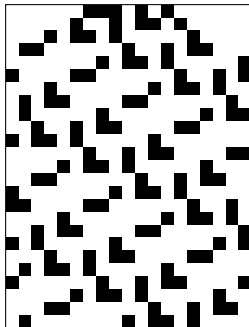
Experimentation

1D

d D

Groups and graphs

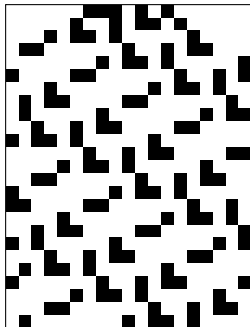
Conclusions



Nope: f_2 has eventually periodic points!

Ville Salo

Experimentation



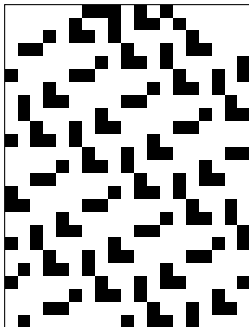
- We notice that

for some $p > 0$:

Nope: f_2 has eventually periodic points!

Ville Salo

Experimentation



- ▶ We notice that

for some $p > 0$:

- Define $f_3 : \{0, 1\}^{\mathbb{Z}} \rightarrow \{0, 1\}^{\mathbb{Z}}$ by
 $f_3(x)_i = 1 \iff x_{[i-3, i+3]} \in K_3$, where

$K_3 = \{0111000, 1000100, 0111100, 1010110, 1111110, 0010001, 0101001, 1101001, 0000101, 0010101, 0011101, 0000011, 1110011, 0011011, 1011011, 0111011, 1111011, 1000111, 1100111, 0010111, 1110111, 0001111, 0111111\}$

- This CA is perhaps nilpotent, but some configurations take 7 steps to die out:



- Define $f_3 : \{0, 1\}^{\mathbb{Z}} \rightarrow \{0, 1\}^{\mathbb{Z}}$ by
 $f_3(x)_i = 1 \iff x_{[i-3, i+3]} \in K_3$, where

$K_3 = \{0111000, 1000100, 0111100, 1010110, 1111110, 0010001, 0101001, 1101001, 0000101, 0010101, 0011101, 0000011, 1110011, 0011011, 1011011, 0111011, 1111011, 1000111, 1100111, 0010111, 1110111, 0001111, 0111111\}$

- This CA is perhaps nilpotent, but some configurations take 7 steps to die out:



- (But maybe it's not nilpotent.)

Busy beavers

- ▶ Nilpotency is undecidable in general, proved in [Aanderaa-Lewis, 1974; Kari, 1992].

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

The questions

Experimentation

1D

d D

Groups and graphs

Conclusions

Busy beavers

- ▶ Nilpotency is undecidable in general, proved in [Aanderaa-Lewis, 1974; Kari, 1992].
- ▶ The maximal nilpotency degree among radius-1 CA on alphabet of size n has busybeaverly growth in n , in the sense that for any computable function $T : \mathbb{N} \rightarrow \mathbb{N}$, there exists n and a radius-1 CA f on $\{0, \dots, n-1\}^{\mathbb{Z}}$ such that

$$T(n) < \inf\{n \mid \forall x \in \{0, \dots, n-1\}^{\mathbb{Z}} : f^n(x) = 0^{\mathbb{Z}}\} < \infty.$$

- ▶ Nilpotency is undecidable in general, proved in [Aanderaa-Lewis, 1974; Kari, 1992].
- ▶ The maximal nilpotency degree among radius-1 CA on alphabet of size n has busybeaverly growth in n , in the sense that for any computable function $T : \mathbb{N} \rightarrow \mathbb{N}$, there exists n and a radius-1 CA f on $\{0, \dots, n-1\}^{\mathbb{Z}}$ such that

$$T(n) < \inf\{n \mid \forall x \in \{0, \dots, n-1\}^{\mathbb{Z}} : f^n(x) = 0^{\mathbb{Z}}\} < \infty.$$

- ▶ Question: Is there some nice simple, memorizable example of a nilpotent CA which has a large nilpotency degree?

- ▶ Nilpotency is undecidable in general, proved in [Aanderaa-Lewis, 1974; Kari, 1992].
- ▶ The maximal nilpotency degree among radius-1 CA on alphabet of size n has busybeaverly growth in n , in the sense that for any computable function $T : \mathbb{N} \rightarrow \mathbb{N}$, there exists n and a radius-1 CA f on $\{0, \dots, n-1\}^{\mathbb{Z}}$ such that

$$T(n) < \inf\{n \mid \forall x \in \{0, \dots, n-1\}^{\mathbb{Z}} : f^n(x) = 0^{\mathbb{Z}}\} < \infty.$$

- ▶ Question: Is there some nice simple, memorable example of a nilpotent CA which has a large nilpotency degree?
- ▶ Question: Among nilpotent binary CA of neighborhood size n , what is the maximal nilpotency degree $MND(n)$ for small n ?

- ▶ Nilpotency is undecidable in general, proved in [Aanderaa-Lewis, 1974; Kari, 1992].
- ▶ The maximal nilpotency degree among radius-1 CA on alphabet of size n has busybeaverly growth in n , in the sense that for any computable function $T : \mathbb{N} \rightarrow \mathbb{N}$, there exists n and a radius-1 CA f on $\{0, \dots, n-1\}^{\mathbb{Z}}$ such that

$$T(n) < \inf\{n \mid \forall x \in \{0, \dots, n-1\}^{\mathbb{Z}} : f^n(x) = 0^{\mathbb{Z}}\} < \infty.$$

- ▶ Question: Is there some nice simple, memorizable example of a nilpotent CA which has a large nilpotency degree?
- ▶ Question: Among nilpotent binary CA of neighborhood size n , what is the maximal nilpotency degree $MND(n)$ for small n ? $MND(3) = 2$, reached by ECA 8.

The settings we look at

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

The questions

Experimentation

1D

d D

Groups and graphs

Conclusions

- ▶ SFTs and sofics in one dimension.

The settings we look at

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

The questions

Experimentation

1D

d D

Groups and graphs

Conclusions

- ▶ SFTs and sofics in one dimension.
- ▶ SFTs (tilings) in d dimensions.

The settings we look at

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

The questions

Experimentation

1D

d D

Groups and graphs

Conclusions

- ▶ SFTs and sofics in one dimension.
- ▶ SFTs (tilings) in d dimensions.
- ▶ Colorings of road-colored graphs / full shifts on groups.

- ▶ Let Σ be a finite alphabet. Then $\Sigma^{\mathbb{Z}}$ is called the *full shift*, and it comes with a natural *shift map*:
 $\sigma(x)_i = x_{i+1}$.

- ▶ Let Σ be a finite alphabet. Then $\Sigma^{\mathbb{Z}}$ is called the *full shift*, and it comes with a natural *shift map*:
 $\sigma(x)_i = x_{i+1}$.
- ▶ A *subshift* is a subset $X \subset \Sigma^{\mathbb{Z}}$ which is shift-invariant ($\sigma(X) = X$) and topologically closed (in the product topology of $\Sigma^{\mathbb{Z}}$).

- ▶ Let Σ be a finite alphabet. Then $\Sigma^{\mathbb{Z}}$ is called the *full shift*, and it comes with a natural *shift map*:
 $\sigma(x)_i = x_{i+1}$.
- ▶ A *subshift* is a subset $X \subset \Sigma^{\mathbb{Z}}$ which is shift-invariant ($\sigma(X) = X$) and topologically closed (in the product topology of $\Sigma^{\mathbb{Z}}$).
- ▶ Equivalent definition: $X \subset \Sigma^{\mathbb{Z}}$ is a subshift if and only if it is defined by a set of forbidden words $F \subset \Sigma^*$ in the sense that for $x \in \Sigma^{\mathbb{Z}}$, we have

$$x \in X \iff \forall w \in F : \forall i \in \mathbb{Z} : x_{[i, i+|w|-1]} \neq w.$$

Languages of subshifts

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

The questions

Experimentation

1D

d D

Groups and graphs

Conclusions

- ▶ If $X \subset \Sigma^{\mathbb{Z}}$ is a subshift, we define its *language* as $\mathcal{L}(X) = \{w \in \Sigma^* \mid \exists x \in X : x_{[0,|w|-1]} = w\}$.

Languages of subshifts

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

The questions

Experimentation

1D

d D

Groups and graphs

Conclusions

- ▶ If $X \subset \Sigma^{\mathbb{Z}}$ is a subshift, we define its *language* as $\mathcal{L}(X) = \{w \in \Sigma^* \mid \exists x \in X : x_{[0,|w|-1]} = w\}$.
- ▶ We also write $w \sqsubset X \iff w \in \mathcal{L}(X)$.

Languages of subshifts

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

The questions

Experimentation

1D

dD

Groups and graphs

Conclusions

- ▶ If $X \subset \Sigma^{\mathbb{Z}}$ is a subshift, we define its *language* as $\mathcal{L}(X) = \{w \in \Sigma^* \mid \exists x \in X : x_{[0,|w|-1]} = w\}$.
- ▶ We also write $w \sqsubset X \iff w \in \mathcal{L}(X)$.
- ▶ If F is a set of forbidden words defining X , then the language of X is the set of words u which can be extended infinitely without introducing a word of F .

Languages of subshifts

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

The questions

Experimentation

1D

dD

Groups and graphs

Conclusions

- ▶ If $X \subset \Sigma^{\mathbb{Z}}$ is a subshift, we define its *language* as $\mathcal{L}(X) = \{w \in \Sigma^* \mid \exists x \in X : x_{[0,|w|-1]} = w\}$.
- ▶ We also write $w \sqsubset X \iff w \in \mathcal{L}(X)$.
- ▶ If F is a set of forbidden words defining X , then the language of X is the set of words u which can be extended infinitely without introducing a word of F .
- ▶ We can also specify a subshift by specifying its language: The language identifies the subshift uniquely.

Types of subshifts: SFTs

- ▶ If $X \subset \Sigma^{\mathbb{Z}}$ can be defined by a finite set of forbidden words, it is a *subshift of finite type* or *SFT*.

Types of subshifts: SFTs

- ▶ If $X \subset \Sigma^{\mathbb{Z}}$ can be defined by a finite set of forbidden words, it is a *subshift of finite type* or *SFT*.
- ▶ For example $\Sigma = \{0, 1\}$ and $F = \{11\}$ define the *golden mean shift* X_λ . For example $01001 \sqsubset X_\lambda$ but $00110 \not\sqsubset X_\lambda$.

Types of subshifts: SFTs

- ▶ If $X \subset \Sigma^{\mathbb{Z}}$ can be defined by a finite set of forbidden words, it is a *subshift of finite type* or *SFT*.
- ▶ For example $\Sigma = \{0, 1\}$ and $F = \{11\}$ define the *golden mean shift* X_λ . For example $01001 \sqsubset X_\lambda$ but $00110 \not\sqsubset X_\lambda$.
- ▶ The golden mean shift is qualitatively similar to a full shift:

Types of subshifts: SFTs

- ▶ If $X \subset \Sigma^{\mathbb{Z}}$ can be defined by a finite set of forbidden words, it is a *subshift of finite type* or *SFT*.
- ▶ For example $\Sigma = \{0, 1\}$ and $F = \{11\}$ define the *golden mean shift* X_λ . For example $01001 \sqsubset X_\lambda$ but $00110 \not\sqsubset X_\lambda$.
- ▶ The golden mean shift is qualitatively similar to a full shift: *topologically transitive* ($u, v \sqsubset X \implies \exists w : uwv \sqsubset X$), and *entropy* (exponential growth rate of the number of words) is positive.

Types of subshifts: SFTs

- ▶ If $X \subset \Sigma^{\mathbb{Z}}$ can be defined by a finite set of forbidden words, it is a *subshift of finite type* or *SFT*.
- ▶ For example $\Sigma = \{0, 1\}$ and $F = \{11\}$ define the *golden mean shift* X_λ . For example $01001 \sqsubset X_\lambda$ but $00110 \not\sqsubset X_\lambda$.
- ▶ The golden mean shift is qualitatively similar to a full shift: *topologically transitive* ($u, v \sqsubset X \implies \exists w : uwv \sqsubset X$), and *entropy* (exponential growth rate of the number of words) is positive.
- ▶ Define $X_\nearrow$ by $\Sigma = \{0, 1\}$ and $F = \{10\}$. For example $000111 \sqsubset X$ but $00110 \not\sqsubset X$.

Types of subshifts: SFTs

- ▶ If $X \subset \Sigma^{\mathbb{Z}}$ can be defined by a finite set of forbidden words, it is a *subshift of finite type* or *SFT*.
- ▶ For example $\Sigma = \{0, 1\}$ and $F = \{11\}$ define the *golden mean shift* X_λ . For example $01001 \sqsubset X_\lambda$ but $00110 \not\sqsubset X_\lambda$.
- ▶ The golden mean shift is qualitatively similar to a full shift: *topologically transitive* ($u, v \sqsubset X \implies \exists w : uwv \sqsubset X$), and *entropy* (exponential growth rate of the number of words) is positive.
- ▶ Define $X_\nearrow$ by $\Sigma = \{0, 1\}$ and $F = \{10\}$. For example $000111 \sqsubset X$ but $00110 \not\sqsubset X$.
- ▶ Qualitatively very different from a full shift:

Types of subshifts: SFTs

- ▶ If $X \subset \Sigma^{\mathbb{Z}}$ can be defined by a finite set of forbidden words, it is a *subshift of finite type* or *SFT*.
- ▶ For example $\Sigma = \{0, 1\}$ and $F = \{11\}$ define the *golden mean shift* X_λ . For example $01001 \sqsubset X_\lambda$ but $00110 \not\sqsubset X_\lambda$.
- ▶ The golden mean shift is qualitatively similar to a full shift: *topologically transitive* ($u, v \sqsubset X \implies \exists w : uwv \sqsubset X$), and *entropy* (exponential growth rate of the number of words) is positive.
- ▶ Define $X_\nearrow$ by $\Sigma = \{0, 1\}$ and $F = \{10\}$. For example $000111 \sqsubset X$ but $00110 \not\sqsubset X$.
- ▶ Qualitatively very different from a full shift: not transitive because 0 cannot occur after 1, and entropy is 0 (because there are $n + 1$ words of length n). This is a *countable SFT*.

Types of subshifts: sofic

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

The questions

Experimentation

1D

dD

Groups and graphs

Conclusions

- ▶ If $X \subset \Sigma^{\mathbb{Z}}$ can be defined by a regular language of forbidden words, it is *sofic*. Equivalently, its language is regular. Equivalently, it is the image of an SFT under a cellular automaton.

Types of subshifts: sofics

- ▶ If $X \subset \Sigma^{\mathbb{Z}}$ can be defined by a regular language of forbidden words, it is *sofic*. Equivalently, its language is regular. Equivalently, it is the image of an SFT under a cellular automaton.
- ▶ For example $\Sigma = \{0, 1\}$, $F = 10(00)^*1 = \{10^n1 \mid n \text{ odd}\}$ defines the *even shift* X_e . For example $1001000011 \sqsubset X$ but $10001 \not\sqsubset X$.

Types of subshifts: sofics

- ▶ If $X \subset \Sigma^{\mathbb{Z}}$ can be defined by a regular language of forbidden words, it is *sofic*. Equivalently, its language is regular. Equivalently, it is the image of an SFT under a cellular automaton.
- ▶ For example $\Sigma = \{0, 1\}$, $F = 10(00)^*1 = \{10^n1 \mid n \text{ odd}\}$ defines the *even shift* X_e . For example $1001000011 \sqsubset X$ but $10001 \not\sqsubset X$.
- ▶ Again qualitatively similar to a full shift (top. transitive, positive entropy), but with a subtle twist: ‘infinite range correlations between cells’.

Types of subshifts: sofics

- ▶ If $X \subset \Sigma^{\mathbb{Z}}$ can be defined by a regular language of forbidden words, it is *sofic*. Equivalently, its language is regular. Equivalently, it is the image of an SFT under a cellular automaton.
- ▶ For example $\Sigma = \{0, 1\}$, $F = 10(00)^*1 = \{10^n1 \mid n \text{ odd}\}$ defines the *even shift* X_e . For example $1001000011 \sqsubset X$ but $10001 \not\sqsubset X$.
- ▶ Again qualitatively similar to a full shift (top. transitive, positive entropy), but with a subtle twist: ‘infinite range correlations between cells’.
- ▶ $F = 10^*1$ defines the *sunny-side-up* subshift $X_{\leq 1}$ where $x \in X_{\leq 1} \iff x = 0^{\mathbb{Z}} \vee x = \dots 00001000\dots$. This is a countable sofic.

- ▶ Let $X \subset \Sigma^{\mathbb{Z}}$ be a subshift. Then $f : X \rightarrow X$ is a *cellular automaton* (CA) if it is continuous and it commutes with the shift: $\sigma \circ f = f \circ \sigma$.

- ▶ Let $X \subset \Sigma^{\mathbb{Z}}$ be a subshift. Then $f : X \rightarrow X$ is a *cellular automaton* (CA) if it is continuous and it commutes with the shift: $\sigma \circ f = f \circ \sigma$.
- ▶ Equivalently, there exist a finite $N \subset \mathbb{Z}$ and $f_{\text{loc}} : \Sigma^N \rightarrow \Sigma$ such that $f(x)_i = f_{\text{loc}}(\sigma^i(x)|_N)$ for all $x \in \Sigma^{\mathbb{Z}}, i \in \mathbb{Z}$.

- ▶ Let $X \subset \Sigma^{\mathbb{Z}}$ be a subshift. Then $f : X \rightarrow X$ is a *cellular automaton* (CA) if it is continuous and it commutes with the shift: $\sigma \circ f = f \circ \sigma$.
- ▶ Equivalently, there exist a finite $N \subset \mathbb{Z}$ and $f_{\text{loc}} : \Sigma^N \rightarrow \Sigma$ such that $f(x)_i = f_{\text{loc}}(\sigma^i(x)|_N)$ for all $x \in \Sigma^{\mathbb{Z}}, i \in \mathbb{Z}$.
- ▶ A map $f : X \rightarrow X$ is a cellular automaton if and only if $f = g|_X$ for some CA $g : \Sigma^{\mathbb{Z}} \rightarrow \Sigma^{\mathbb{Z}}$ (because we can fill in the local rule arbitrarily).

CA on one-dimensional full shifts

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

The questions

Experimentation

1D

*d*D

Groups and graphs

Conclusions

- ▶ We already saw examples of CA on $\{0, 1\}^{\mathbb{Z}}$.

An example of CA on golden mean shift

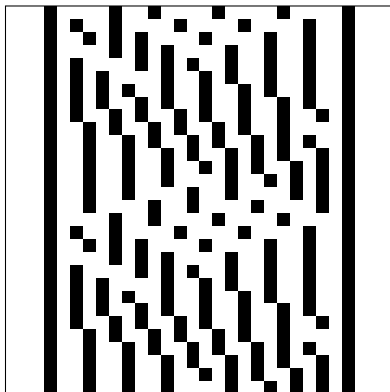
- Define $g : X_\lambda \rightarrow X_\lambda$ by exchanging $100001 \leftrightarrow 101001$ everywhere and h by $100001 \leftrightarrow 100101$. Let $f = h \circ g$.

An example of CA on golden mean shift

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

- Define $g : X_\lambda \rightarrow X_\lambda$ by exchanging $100001 \leftrightarrow 101001$ everywhere and h by $100001 \leftrightarrow 100101$. Let $f = h \circ g$. Here's f on input
...000100001001000010000100001000....



The questions

Experimentation

1D

dD

Groups and graphs

Conclusions

Examples of CA on countable SFTs

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

The questions

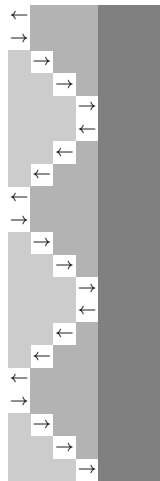
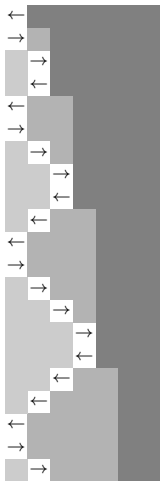
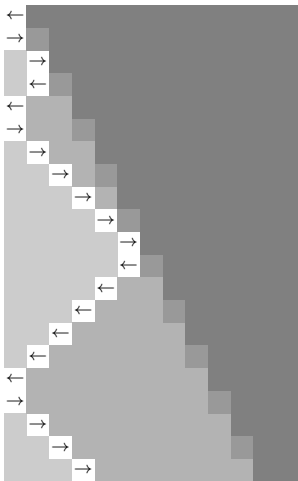
Experimentation

1D

dD

Groups and graphs

Conclusions



Examples of CA on uncountable sofic shifts

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

The questions

Experimentation

1D

dD

Groups and graphs

Conclusions

- ▶ Any CA can be simulated on any uncountable sofic shift:

Theorem

Let $f : A^{\mathbb{Z}} \rightarrow A^{\mathbb{Z}}$ be a CA, and $X \subset \Sigma^{\mathbb{Z}}$ an uncountable sofic shift. Then there is $n \in \mathbb{N}$, a function $\tau : A \rightarrow \Sigma^n$ and a CA $g : X \rightarrow X$ such that $g(\tau(x)) = \tau(f(x))$ for all $x \in A^{\mathbb{Z}}$.

Example of CA on countable sofic shift

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

The questions

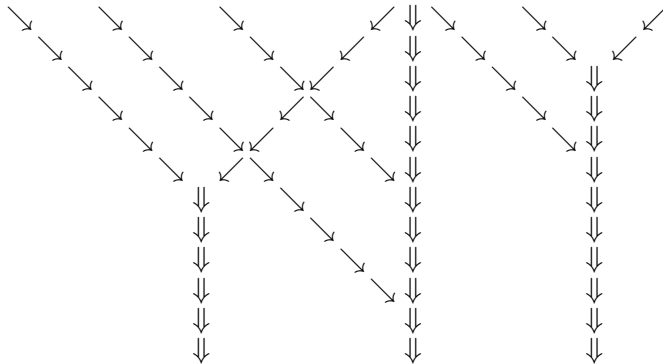
Experimentation

1D

dD

Groups and graphs

Conclusions



- ▶ A *pointed subshift* is a pair (X, x) where $X \subset \Sigma^{\mathbb{Z}}$ is a subshift and $x = a^{\mathbb{Z}}$ for some $a \in \Sigma$. Usually $x = 0^{\mathbb{Z}}$, and we write 0 for this point, and do not mention it explicitly.

- ▶ A *pointed subshift* is a pair (X, x) where $X \subset \Sigma^{\mathbb{Z}}$ is a subshift and $x = a^{\mathbb{Z}}$ for some $a \in \Sigma$. Usually $x = 0^{\mathbb{Z}}$, and we write 0 for this point, and do not mention it explicitly.
- ▶ Suppose now that X is pointed and $f : X \rightarrow X$ is a CA.

Nilpotency

- ▶ A *pointed subshift* is a pair (X, x) where $X \subset \Sigma^{\mathbb{Z}}$ is a subshift and $x = a^{\mathbb{Z}}$ for some $a \in \Sigma$. Usually $x = 0^{\mathbb{Z}}$, and we write 0 for this point, and do not mention it explicitly.
- ▶ Suppose now that X is pointed and $f : X \rightarrow X$ is a CA.
- ▶ f is *nilpotent* if $\exists n : \forall x \in X : f^n(x) = 0$.

- ▶ A *pointed subshift* is a pair (X, x) where $X \subset \Sigma^{\mathbb{Z}}$ is a subshift and $x = a^{\mathbb{Z}}$ for some $a \in \Sigma$. Usually $x = 0^{\mathbb{Z}}$, and we write 0 for this point, and do not mention it explicitly.
- ▶ Suppose now that X is pointed and $f : X \rightarrow X$ is a CA.
- ▶ f is *nilpotent* if $\exists n : \forall x \in X : f^n(x) = 0$.
- ▶ f is *weakly nilpotent* if $\forall x \in X : \exists n : f^n(x) = 0$.

- ▶ A *pointed subshift* is a pair (X, x) where $X \subset \Sigma^{\mathbb{Z}}$ is a subshift and $x = a^{\mathbb{Z}}$ for some $a \in \Sigma$. Usually $x = 0^{\mathbb{Z}}$, and we write 0 for this point, and do not mention it explicitly.
- ▶ Suppose now that X is pointed and $f : X \rightarrow X$ is a CA.
- ▶ f is *nilpotent* if $\exists n : \forall x \in X : f^n(x) = 0$.
- ▶ f is *weakly nilpotent* if $\forall x \in X : \exists n : f^n(x) = 0$.
- ▶ f is *asymptotically nilpotent* (AN) if $\forall x \in X : f^n(x) \rightarrow 0$.

- ▶ A *pointed subshift* is a pair (X, x) where $X \subset \Sigma^{\mathbb{Z}}$ is a subshift and $x = a^{\mathbb{Z}}$ for some $a \in \Sigma$. Usually $x = 0^{\mathbb{Z}}$, and we write 0 for this point, and do not mention it explicitly.
- ▶ Suppose now that X is pointed and $f : X \rightarrow X$ is a CA.
- ▶ f is *nilpotent* if $\exists n : \forall x \in X : f^n(x) = 0$.
- ▶ f is *weakly nilpotent* if $\forall x \in X : \exists n : f^n(x) = 0$.
- ▶ f is *asymptotically nilpotent* (AN) if $\forall x \in X : f^n(x) \rightarrow 0$.
- ▶ f is *uniformly asymptotically nilpotent* (UAN) if $f^n(x) \rightarrow 0$ uniformly in x .

Theorem

Let $X \subset \Sigma^{\mathbb{Z}}$ be a pointed subshift. A CA $f : X \rightarrow X$ is nilpotent iff weakly nilpotent iff uniformly asymptotically nilpotent. Nilpotency implies asymptotic nilpotency.

The questions

Experimentation

1D

dD

Groups and graphs

Conclusions

Theorem

Let $X \subset \Sigma^{\mathbb{Z}}$ be a pointed subshift. A CA $f : X \rightarrow X$ is nilpotent iff weakly nilpotent iff uniformly asymptotically nilpotent. Nilpotency implies asymptotic nilpotency.

- Only question left: Does asymptotic nilpotency imply nilpotency?

The questions

Experimentation

1D

dD

Groups and graphs

Conclusions

Theorem

Let $X \subset \Sigma^{\mathbb{Z}}$ be a pointed subshift. A CA $f : X \rightarrow X$ is nilpotent iff weakly nilpotent iff uniformly asymptotically nilpotent. Nilpotency implies asymptotic nilpotency.

- ▶ Only question left: Does asymptotic nilpotency imply nilpotency?
- ▶ We say f is *strictly asymptotically nilpotent* (SAN) if it is nilpotent but not asymptotically nilpotent.

The questions

Experimentation

1D

dD

Groups and graphs

Conclusions

Theorem

Let $X \subset \Sigma^{\mathbb{Z}}$ be a pointed subshift. A CA $f : X \rightarrow X$ is nilpotent iff weakly nilpotent iff uniformly asymptotically nilpotent. Nilpotency implies asymptotic nilpotency.

- ▶ Only question left: Does asymptotic nilpotency imply nilpotency?
- ▶ We say f is *strictly asymptotically nilpotent* (SAN) if it is nilpotent but not asymptotically nilpotent.
- ▶ We say that a subshift is *nilrigid* if it does not support any cellular automata that are SAN. We say that a group G is *nilrigid* if Σ^G is nilrigid for all Σ .

The questions

Experimentation

1D

dD

Groups and graphs

Conclusions

$\mathbb{Z}$ is nilrigid

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

The questions

Experimentation

1D

$d\mathbb{D}$

Groups and graphs

Conclusions

Theorem (Guillon-Richard, '10)

Let $f : \Sigma^{\mathbb{Z}} \rightarrow \Sigma^{\mathbb{Z}}$ be asymptotically nilpotent. Then f is nilpotent.

Theorem (Guillon-Richard, '10)

Let $f : \Sigma^{\mathbb{Z}} \rightarrow \Sigma^{\mathbb{Z}}$ be asymptotically nilpotent. Then f is nilpotent.

- Proof idea: AN implies that some pattern w must forces an infinite column of 0s. Use w to trick the CA.

Theorem (Guillon-Richard, '10)

Let $f : \Sigma^{\mathbb{Z}} \rightarrow \Sigma^{\mathbb{Z}}$ be asymptotically nilpotent. Then f is nilpotent.

- ▶ Proof idea: AN implies that some pattern w must forces an infinite column of 0s. Use w to trick the CA.
- ▶ Alternative (more general) result is proved in [Saló, '16] based on studying the subshift of all spacetime diagrams. (And almost follows from [Pavlov-Schraudner, '15].)

$\mathbb{Z}$ -SFTs are nilrigid

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

Theorem (Guillon-Richard, '10)

Let $X \subset \Sigma^{\mathbb{Z}}$ be a transitive SFT, and let $f : X \rightarrow X$ be an asymptotically nilpotent CA. Then f is nilpotent.

The questions

Experimentation

1D

dD

Groups and graphs

Conclusions

$\mathbb{Z}$ -SFTs are nilrigid

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

Theorem (Guillon-Richard, '10)

Let $X \subset \Sigma^{\mathbb{Z}}$ be a transitive SFT, and let $f : X \rightarrow X$ be an asymptotically nilpotent CA. Then f is nilpotent.

- Proof idea: Same as on a full shift.

The questions

Experimentation

1D

$d\mathbb{D}$

Groups and graphs

Conclusions

$\mathbb{Z}$ -SFTs are nilrigid

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

Theorem (Guillon-Richard, '10)

Let $X \subset \Sigma^{\mathbb{Z}}$ be a transitive SFT, and let $f : X \rightarrow X$ be an asymptotically nilpotent CA. Then f is nilpotent.

- Proof idea: Same as on a full shift.

Theorem (Salo, '12)

Let $X \subset \Sigma^{\mathbb{Z}}$ be an SFT, and let $f : X \rightarrow X$ be an asymptotically nilpotent CA. Then f is nilpotent.

The questions

Experimentation

1D

dD

Groups and graphs

Conclusions

$\mathbb{Z}$ -SFTs are nilrigid

Theorem (Guillon-Richard, '10)

Let $X \subset \Sigma^{\mathbb{Z}}$ be a transitive SFT, and let $f : X \rightarrow X$ be an asymptotically nilpotent CA. Then f is nilpotent.

- ▶ Proof idea: Same as on a full shift.

Theorem (Salo, '12)

Let $X \subset \Sigma^{\mathbb{Z}}$ be an SFT, and let $f : X \rightarrow X$ be an asymptotically nilpotent CA. Then f is nilpotent.

- ▶ Only sketched in [Salo, '12]. Proof idea: CA on SFTs map transitive components to (possibly other) transitive components. On any cycle, the CA is nilpotent by [Guillon-Richard, '10].

Not all $\mathbb{Z}$ -sofics are nilrigid

- ▶ The sunny-side-up $X_{\leq 1}$ has a SAN CA: the shift map σ ! $X_{\leq 1}$ is countable.

Not all $\mathbb{Z}$ -sofics are nilrigid

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

The questions

Experimentation

1D

dD

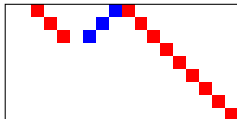
Groups and graphs

Conclusions

- ▶ The sunny-side-up $X_{\leq 1}$ has a SAN CA: the shift map σ ! $X_{\leq 1}$ is countable.
- ▶ Define $X \subset \{0, 1, 2\}^{\mathbb{Z}}$ with forbidden language $10^*1 + 20^*2$ (so 1s and 2s alternate). It is topologically transitive and has positive entropy.

Not all $\mathbb{Z}$ -sofics are nilrigid

- ▶ The sunny-side-up $X_{\leq 1}$ has a SAN CA: the shift map σ ! $X_{\leq 1}$ is countable.
- ▶ Define $X \subset \{0, 1, 2\}^{\mathbb{Z}}$ with forbidden language $10^*1 + 20^*2$ (so 1s and 2s alternate). It is topologically transitive and has positive entropy. Let f be a CA that moves 1s left and 2s right, erasing both on collision. This CA is SAN.



Which $\mathbb{Z}$ -sofics are nilrigid?

Question

Which one-dimensional sofic shifts are nilrigid, that is, do not support cellular automata which are SAN?

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

The questions

Experimentation

1D

dD

Groups and graphs

Conclusions

Which $\mathbb{Z}$ -sofics are nilrigid?

Question

Which one-dimensional sofic shifts are nilrigid, that is, do not support cellular automata which are SAN?

- ▶ To recap.... SFTs are nilrigid.

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

The questions

Experimentation

1D

dD

Groups and graphs

Conclusions

Which $\mathbb{Z}$ -sofics are nilrigid?

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

Question

Which one-dimensional sofic shifts are nilrigid, that is, do not support cellular automata which are SAN?

The questions

Experimentation

1D

dD

Groups and graphs

Conclusions

- ▶ To recap.... SFTs are nilrigid.
- ▶ The sunny-side-up and the ternary sofic X with forbidden language $10^*1 + 20^*2$ are not.

Which $\mathbb{Z}$ -sofics are nilrigid?

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

Question

Which one-dimensional sofic shifts are nilrigid, that is, do not support cellular automata which are SAN?

The questions

Experimentation

1D

dD

Groups and graphs

Conclusions

- ▶ To recap.... SFTs are nilrigid.
- ▶ The sunny-side-up and the ternary sofic X with forbidden language $10^*1 + 20^*2$ are not.
- ▶ The binary SFT with forbidden language 01 is nilrigid, even though similar to the sunny-side-up (both are very simple countable subshifts).

Which $\mathbb{Z}$ -sofics are nilrigid?

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

Question

Which one-dimensional sofic shifts are nilrigid, that is, do not support cellular automata which are SAN?

The questions

Experimentation

1D

dD

Groups and graphs

Conclusions

- ▶ To recap.... SFTs are nilrigid.
- ▶ The sunny-side-up and the ternary sofic X with forbidden language $10^*1 + 20^*2$ are not.
- ▶ The binary SFT with forbidden language 01 is nilrigid, even though similar to the sunny-side-up (both are very simple countable subshifts).
- ▶ The binary sofic X_e with forbidden language $10(00)^*1$ is nilrigid, even though similar to X above (both are top. transitive and have positive entropy).

Which $\mathbb{Z}$ -sofics are nilrigid?

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

Question

Which one-dimensional sofic shifts are nilrigid, that is, do not support cellular automata which are SAN?

The questions

Experimentation

1D

dD

Groups and graphs

Conclusions

- ▶ To recap.... SFTs are nilrigid.
- ▶ The sunny-side-up and the ternary sofic X with forbidden language $10^*1 + 20^*2$ are not.
- ▶ The binary SFT with forbidden language 01 is nilrigid, even though similar to the sunny-side-up (both are very simple countable subshifts).
- ▶ The binary sofic X_e with forbidden language $10(00)^*1$ is nilrigid, even though similar to X above (both are top. transitive and have positive entropy).
- ▶ Is there a simple characterization? More generally, which dynamical properties forbid the construction of SAN CA?

Higher dimensions

- ▶ We can generalize subshifts to higher dimensions: $\Sigma^{\mathbb{Z}^d}$ is called a full shift, and comes up with d shift maps $\sigma_i(x)_{\vec{v}} = x_{\vec{v}+e_i}$ where e_i are a basis of $\mathbb{Z}^d$.

Higher dimensions

- ▶ We can generalize subshifts to higher dimensions: $\Sigma^{\mathbb{Z}^d}$ is called a full shift, and comes up with d shift maps $\sigma_i(x)_{\vec{v}} = x_{\vec{v}+e_i}$ where e_i are a basis of $\mathbb{Z}^d$.
- ▶ Again subshifts are topologically closed sets $X \subset \Sigma^{\mathbb{Z}^d}$ which are invariant under the shift maps σ_i . Again equivalently defined by forbidden patterns $P \in \Sigma^D$ where $D \subset \mathbb{Z}^d$ is finite.

- ▶ We can generalize subshifts to higher dimensions: $\Sigma^{\mathbb{Z}^d}$ is called a full shift, and comes up with d shift maps $\sigma_i(x)_{\vec{v}} = x_{\vec{v}+e_i}$ where e_i are a basis of $\mathbb{Z}^d$.
- ▶ Again subshifts are topologically closed sets $X \subset \Sigma^{\mathbb{Z}^d}$ which are invariant under the shift maps σ_i . Again equivalently defined by forbidden patterns $P \in \Sigma^D$ where $D \subset \mathbb{Z}^d$ is finite.
- ▶ An SFT is defined by finitely many forbidden patterns.

- ▶ We can generalize subshifts to higher dimensions: $\Sigma^{\mathbb{Z}^d}$ is called a full shift, and comes up with d shift maps $\sigma_i(x)_{\vec{v}} = x_{\vec{v}+e_i}$ where e_i are a basis of $\mathbb{Z}^d$.
- ▶ Again subshifts are topologically closed sets $X \subset \Sigma^{\mathbb{Z}^d}$ which are invariant under the shift maps σ_i . Again equivalently defined by forbidden patterns $P \in \Sigma^D$ where $D \subset \mathbb{Z}^d$ is finite.
- ▶ An SFT is defined by finitely many forbidden patterns.
- ▶ In one dimension, SFTs can be understood well through automata theory. In two dimensions (and up), Turing machines can be simulated, so we step into the realm of recursion theory.

Multidimensional CA

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

The questions

Experimentation

1D

d D

Groups and graphs

Conclusions

- ▶ Same definition as before.

Multidimensional CA

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

The questions

Experimentation

1D

*d*D

Groups and graphs

Conclusions

- ▶ Same definition as before.
- ▶ Let $X \subset \Sigma^{\mathbb{Z}^d}$ be a subshift. Then $f : X \rightarrow X$ is a *cellular automaton* (CA) if it is continuous and it commutes with the shifts: $\sigma_i \circ f = f \circ \sigma_i$.

- ▶ Same definition as before.
- ▶ Let $X \subset \Sigma^{\mathbb{Z}^d}$ be a subshift. Then $f : X \rightarrow X$ is a *cellular automaton* (CA) if it is continuous and it commutes with the shifts: $\sigma_i \circ f = f \circ \sigma_i$.
- ▶ Equivalently, there exist a finite $N \subset \mathbb{Z}^d$ and $f_{\text{loc}} : \Sigma^N \rightarrow \Sigma$ such that $f(x)_{\vec{v}} = f_{\text{loc}}(\sigma_{\vec{v}}(x)|_N)$ for all $x \in X, \vec{v} \in \mathbb{Z}^d$.

- ▶ Same definition as before.
- ▶ Let $X \subset \Sigma^{\mathbb{Z}^d}$ be a subshift. Then $f : X \rightarrow X$ is a *cellular automaton* (CA) if it is continuous and it commutes with the shifts: $\sigma_i \circ f = f \circ \sigma_i$.
- ▶ Equivalently, there exist a finite $N \subset \mathbb{Z}^d$ and $f_{\text{loc}} : \Sigma^N \rightarrow \Sigma$ such that $f(x)_{\vec{v}} = f_{\text{loc}}(\sigma_{\vec{v}}(x)|_N)$ for all $x \in X, \vec{v} \in \mathbb{Z}^d$.
- ▶ A map $f : X \rightarrow X$ is a cellular automaton if and only if $f = g|_X$ for some CA $g : \Sigma^{\mathbb{Z}^d} \rightarrow \Sigma^{\mathbb{Z}^d}$ (because we can fill in the local rule arbitrarily).

CA on multidimensional SFTs

- ▶ On 2D full shifts, an example is Game of Life.

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

The questions

Experimentation

1D

dD

Groups and graphs

Conclusions

CA on multidimensional SFTs

- ▶ On 2D full shifts, an example is Game of Life.
- ▶ Even one-step properties of CA on two-dimensional SFTs are often undecidable (surjectivity and injectivity for example).

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

The questions

Experimentation

1D

dD

Groups and graphs

Conclusions

CA on multidimensional SFTs

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

The questions

Experimentation

1D

dD

Groups and graphs

Conclusions

- ▶ On 2D full shifts, an example is Game of Life.
- ▶ Even one-step properties of CA on two-dimensional SFTs are often undecidable (surjectivity and injectivity for example).
- ▶ If $X \subset \Sigma^{\mathbb{Z}^d}$ is an SFT with positive entropy, then there are many patterns with the same borders. This allows the construction of nontrivial cellular automata. (e.g. [Hochman, '09])

CA on multidimensional SFTs

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

The questions

Experimentation

1D

dD

Groups and graphs

Conclusions

- ▶ On 2D full shifts, an example is Game of Life.
- ▶ Even one-step properties of CA on two-dimensional SFTs are often undecidable (surjectivity and injectivity for example).
- ▶ If $X \subset \Sigma^{\mathbb{Z}^d}$ is an SFT with positive entropy, then there are many patterns with the same borders. This allows the construction of nontrivial cellular automata. (e.g. [Hochman, '09])
- ▶ If $X \subset \Sigma^{\mathbb{Z}^d}$ is an SFT and $f : X \rightarrow X$ is a CA, then there is an SFT $Y \subset \Sigma^{\mathbb{Z}^{d+1}}$ such that $\sigma_{d+1} : Y \rightarrow Y$ is (conjugate to the injectivization of) f on its limit set.

CA on multidimensional SFTs

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

The questions

Experimentation

1D

*d*D

Groups and graphs

Conclusions

- ▶ On 2D full shifts, an example is Game of Life.
- ▶ Even one-step properties of CA on two-dimensional SFTs are often undecidable (surjectivity and injectivity for example).
- ▶ If $X \subset \Sigma^{\mathbb{Z}^d}$ is an SFT with positive entropy, then there are many patterns with the same borders. This allows the construction of nontrivial cellular automata. (e.g. [Hochman, '09])
- ▶ If $X \subset \Sigma^{\mathbb{Z}^d}$ is an SFT and $f : X \rightarrow X$ is a CA, then there is an SFT $Y \subset \Sigma^{\mathbb{Z}^{d+1}}$ such that $\sigma_{d+1} : Y \rightarrow Y$ is (conjugate to the injectivization of) f on its limit set.
- ▶ Let $X \subset \Sigma^{\mathbb{Z}^d}$ be any subshift. Then the shift maps σ_i are cellular automata (clearly they commute with shift maps). Quite often these are the *only* CA on a given subshift, even on SFTs when $d \geq 2$.

Abelian groups are nilrigid

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

Theorem (S., '12)

Let $f : \Sigma^{\mathbb{Z}^d} \rightarrow \Sigma^{\mathbb{Z}^d}$ be asymptotically nilpotent. Then f is nilpotent.

The questions

Experimentation

1D

d D

Groups and graphs

Conclusions

Abelian groups are nilrigid

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

Theorem (S., '12)

Let $f : \Sigma^{\mathbb{Z}^d} \rightarrow \Sigma^{\mathbb{Z}^d}$ be asymptotically nilpotent. Then f is nilpotent.

- ▶ Proof idea: Reduce to lower dimension by periodization. Use this to prove that information cannot flow in any dimension for more than a bounded number of steps.

The questions

Experimentation

1D

dD

Groups and graphs

Conclusions

Abelian groups are nilrigid

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

Theorem (S., '12)

Let $f : \Sigma^{\mathbb{Z}^d} \rightarrow \Sigma^{\mathbb{Z}^d}$ be asymptotically nilpotent. Then f is nilpotent.

- Proof idea: Reduce to lower dimension by periodization. Use this to prove that information cannot flow in any dimension for more than a bounded number of steps.

Theorem (S., '12)

Let $f : X \rightarrow X$ be asymptotically nilpotent, where $X \subset \Sigma^{\mathbb{Z}^d}$ is an SFT with dense finite points. Then f is nilpotent.

The questions

Experimentation

1D

dD

Groups and graphs

Conclusions

Abelian groups are nilrigid

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

The questions

Experimentation

1D

dD

Groups and graphs

Conclusions

Theorem (S., '12)

Let $f : \Sigma^{\mathbb{Z}^d} \rightarrow \Sigma^{\mathbb{Z}^d}$ be asymptotically nilpotent. Then f is nilpotent.

- Proof idea: Reduce to lower dimension by periodization. Use this to prove that information cannot flow in any dimension for more than a bounded number of steps.

Theorem (S., '12)

Let $f : X \rightarrow X$ be asymptotically nilpotent, where $X \subset \Sigma^{\mathbb{Z}^d}$ is an SFT with dense finite points. Then f is nilpotent.

Theorem (S., '16)

Let $X \subset \Sigma^{\mathbb{Z}^2}$ be an SFT. Then no shift map is SAN.

Which multidimensional SFTs are nilrigid?

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

Question

Which multidimensional SFTs support cellular automata that are asymptotically nilpotent but not nilpotent? Do any?

The questions

Experimentation

1D

*d*D

Groups and graphs

Conclusions

Which multidimensional SFTs are nilrigid?

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

Question

Which multidimensional SFTs support cellular automata that are asymptotically nilpotent but not nilpotent? Do any?

Conjecture

There is a three-dimensional SFT $X \subset \Sigma^{\mathbb{Z}^3}$ such that σ_1 is SAN.

The questions

Experimentation

1D

d D

Groups and graphs

Conclusions

Which multidimensional SFTs are nilrigid?

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

Question

Which multidimensional SFTs support cellular automata that are asymptotically nilpotent but not nilpotent? Do any?

Conjecture

There is a three-dimensional SFT $X \subset \Sigma^{\mathbb{Z}^3}$ such that σ_1 is SAN.

- There seems to be such an SFT, work in progress with Ilkka Törmä.

The questions

Experimentation

1D

d D

Groups and graphs

Conclusions

Which multidimensional SFTs are nilrigid?

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

The questions

Experimentation

1D

d D

Groups and graphs

Conclusions

Question

Which multidimensional SFTs support cellular automata that are asymptotically nilpotent but not nilpotent? Do any?

Conjecture

There is a three-dimensional SFT $X \subset \Sigma^{\mathbb{Z}^3}$ such that σ_1 is SAN.

- ▶ There seems to be such an SFT, work in progress with Ilkka Törmä.
- ▶ Are there SAN CA on $\mathbb{Z}^2$ -SFTs?

Full shift on a group

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

The questions

Experimentation

1D

dD

Groups and graphs

Conclusions

- ▶ Let $(G, \cdot)$ be any finitely generated group. Then Σ^G is called the *full shift* on G . We have an action of G on it by $\sigma_g(x)_h = x_{g^{-1}h}$ for $x \in \Sigma^G$ and $g, h \in G$.

Full shift on a group

- ▶ Let $(G, \cdot)$ be any finitely generated group. Then Σ^G is called the *full shift* on G . We have an action of G on it by $\sigma_g(x)_h = x_{g^{-1}h}$ for $x \in \Sigma^G$ and $g, h \in G$.
- ▶ Groups can be seen as labeled directed graphs: for example $\mathbb{Z}^d$ can be seen as $(\mathbb{Z}^d, \{(\vec{v}, \vec{v}') \mid \vec{v}' - \vec{v} \in \{e_1, \dots, e_d, -e_1, \dots, -e_d\}\})$. We can color an edge $(\vec{v}, \vec{v}')$ by the offset $\vec{v}' - \vec{v}$ to get a labeled graph.

Full shift on a group

- ▶ Let $(G, \cdot)$ be any finitely generated group. Then Σ^G is called the *full shift* on G . We have an action of G on it by $\sigma_g(x)_h = x_{g^{-1}h}$ for $x \in \Sigma^G$ and $g, h \in G$.
- ▶ Groups can be seen as labeled directed graphs: for example $\mathbb{Z}^d$ can be seen as $(\mathbb{Z}^d, \{(\vec{v}, \vec{v}') \mid \vec{v}' - \vec{v} \in \{e_1, \dots, e_d, -e_1, \dots, -e_d\}\})$. We can color an edge $(\vec{v}, \vec{v}')$ by the offset $\vec{v}' - \vec{v}$ to get a labeled graph.
- ▶ More generally, let S be any finite generating set for G . We define the *Cayley graph* of the group (with respect to S) to be the labeled graph (G, E, λ, S) where $\lambda : E \rightarrow S$ and $(g, h) \in E \iff h = gs$ for some $s \in S$, and $\lambda(g, gs) = s$ for all $g \in G$.

Full shifts on general graphs

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

The questions

Experimentation

1D

d D

Groups and graphs

Conclusions

- ▶ Let $G = (V, E, \lambda, S)$ be a directed *road-colored* graph: the edges are colored by $\lambda : E \rightarrow S$ (S finite) in such a way that for every vertex $v \in V$, for each $s \in S$ there is a unique edge (v, v') with $\lambda(v, v') = s$.

Full shifts on general graphs

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

The questions

Experimentation

1D

d D

Groups and graphs

Conclusions

- ▶ Let $G = (V, E, \lambda, S)$ be a directed *road-colored* graph: the edges are colored by $\lambda : E \rightarrow S$ (S finite) in such a way that for every vertex $v \in V$, for each $s \in S$ there is a unique edge (v, v') with $\lambda(v, v') = s$.
- ▶ The terminal vertex v' of the unique edge from v with label $s \in S$ is denoted by vs . If $u \in S^*$, we write $vu = (\cdots ((vu_0)u_1) \cdots u_{|u|-1})$. Write Σ^G for the set of colorings of the nodes of G .

Graphs as groups

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

- ▶ The nicest setting for cellular automata are graphs which look the same at every vertex, that is, ones where the automorphism group acts vertex-transitively.

The questions

Experimentation

1D

dD

Groups and graphs

Conclusions

- ▶ The nicest setting for cellular automata are graphs which look the same at every vertex, that is, ones where the automorphism group acts vertex-transitively.

Lemma (Variant of Sabidussi theorem)

If $G = (V, E, \lambda, S)$ is directed, weakly connected, the automorphism group acts vertex-transitively, and G and G^R are road-colored, then for any $v \in V$, there is a canonical group structure $(V, \cdot, v)$ with v as the identity element and with some generating set $\{\bar{s} \mid s \in S\} \subset V$ such that

$$E = \{(v, v \cdot \bar{s}) \mid v \in V, s \in S\} \text{ and } \lambda(v, v \cdot \bar{s}) = s$$

that is, G is the Cayley graph of $(V, \cdot, v)$ with respect to $\bar{S}$.

CA on groups

- ▶ Same definition as before.

CA on groups

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

The questions

Experimentation

1D

d D

Groups and graphs

Conclusions

- ▶ Same definition as before.
- ▶ A map $f : \Sigma^G \rightarrow \Sigma^G$ is a *cellular automaton* (CA) if it is continuous and it commutes with the shifts:
$$\sigma_g \circ f = f \circ \sigma_g.$$

CA on groups

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

The questions

Experimentation

1D

dD

Groups and graphs

Conclusions

- ▶ Same definition as before.
- ▶ A map $f : \Sigma^G \rightarrow \Sigma^G$ is a *cellular automaton* (CA) if it is continuous and it commutes with the shifts:
$$\sigma_g \circ f = f \circ \sigma_g.$$
- ▶ Equivalently, there exist a finite $N \subset G$ and $f_{\text{loc}} : \Sigma^N \rightarrow \Sigma$ such that $f(x)_g = f_{\text{loc}}(\sigma_{g^{-1}}(x)|_N)$ for all $x \in \Sigma^G, g \in G$.

CA on groups

- ▶ Same definition as before.
- ▶ A map $f : \Sigma^G \rightarrow \Sigma^G$ is a *cellular automaton* (CA) if it is continuous and it commutes with the shifts:
$$\sigma_g \circ f = f \circ \sigma_g.$$
- ▶ Equivalently, there exist a finite $N \subset G$ and $f_{\text{loc}} : \Sigma^N \rightarrow \Sigma$ such that $f(x)_g = f_{\text{loc}}(\sigma_{g^{-1}}(x)|_N)$ for all $x \in \Sigma^G, g \in G$.
- ▶ The first definition above also works on subshifts on groups. The second works on all road-colored graphs.

CA on groups

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

The questions

Experimentation

1D

dD

Groups and graphs

Conclusions

- ▶ Same definition as before.
- ▶ A map $f : \Sigma^G \rightarrow \Sigma^G$ is a *cellular automaton* (CA) if it is continuous and it commutes with the shifts:
$$\sigma_g \circ f = f \circ \sigma_g.$$
- ▶ Equivalently, there exist a finite $N \subset G$ and $f_{\text{loc}} : \Sigma^N \rightarrow \Sigma$ such that $f(x)_g = f_{\text{loc}}(\sigma_{g^{-1}}(x)|_N)$ for all $x \in \Sigma^G, g \in G$.
- ▶ The first definition above also works on subshifts on groups. The second works on all road-colored graphs.
- ▶ On non-commutative groups, one must be careful with left and right multiplication – but local rules just look at ‘neighbors in the Cayley graph’.

- ▶ Same definition as before.
- ▶ A map $f : \Sigma^G \rightarrow \Sigma^G$ is a *cellular automaton* (CA) if it is continuous and it commutes with the shifts:
$$\sigma_g \circ f = f \circ \sigma_g.$$
- ▶ Equivalently, there exist a finite $N \subset G$ and $f_{\text{loc}} : \Sigma^N \rightarrow \Sigma$ such that $f(x)_g = f_{\text{loc}}(\sigma_{g^{-1}}(x)|_N)$ for all $x \in \Sigma^G, g \in G$.
- ▶ The first definition above also works on subshifts on groups. The second works on all road-colored graphs.
- ▶ On non-commutative groups, one must be careful with left and right multiplication – but local rules just look at ‘neighbors in the Cayley graph’.
- ▶ New value of x_g is determined by the values x_{gh} at finitely many $h \in N$.

Notes on CA on groups

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

The questions

Experimentation

1D

d D

Groups and graphs

Conclusions

- ▶ Let $X = \Sigma^G$ where G is any group. Then the 'right-shifts' $\sigma_g^R(x)_h = x_{hg}$ are cellular automata, but the actual shift maps σ_g are not (if g is not in the center).

Notes on CA on groups

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

The questions

Experimentation

1D

d D

Groups and graphs

Conclusions

- ▶ Let $X = \Sigma^G$ where G is any group. Then the 'right-shifts' $\sigma_g^R(x)_h = x_{hg}$ are cellular automata, but the actual shift maps σ_g are not (if g is not in the center).
- ▶ There are plenty of CA on full shifts Σ^G (where G is any group or road-colored graph): just pick any local rule.

Notes on CA on groups

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

The questions

Experimentation

1D

d D

Groups and graphs

Conclusions

- ▶ Let $X = \Sigma^G$ where G is any group. Then the 'right-shifts' $\sigma_g^R(x)_h = x_{hg}$ are cellular automata, but the actual shift maps σ_g are not (if g is not in the center).
- ▶ There are plenty of CA on full shifts Σ^G (where G is any group or road-colored graph): just pick any local rule.
- ▶ On the free group, there is a subshift that supports no cellular automata except the identity function.

Notes on CA on groups

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

The questions

Experimentation

1D

d D

Groups and graphs

Conclusions

- ▶ Let $X = \Sigma^G$ where G is any group. Then the 'right-shifts' $\sigma_g^R(x)_h = x_{hg}$ are cellular automata, but the actual shift maps σ_g are not (if g is not in the center).
- ▶ There are plenty of CA on full shifts Σ^G (where G is any group or road-colored graph): just pick any local rule.
- ▶ On the free group, there is a subshift that supports no cellular automata except the identity function.
- ▶ Nilpotency is undecidable on any nontrivial full shift on any finitely generated infinite group.

SAN CA on road-colored graphs

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

The questions

Experimentation

1D

d D

Groups and graphs

Conclusions

- ▶ Let $f : \Sigma^{\mathbb{Z}} \rightarrow \Sigma^{\mathbb{Z}}$ be a CA that is not nilpotent but is nilpotent on periodic points. Take a graph G consisting of copies of cycle graphs C_n . Then the local rule of f induces a CA on colorings of G , and this CA is SAN.

SAN CA on road-colored graphs

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

The questions

Experimentation

1D

d D

Groups and graphs

Conclusions

- ▶ Let $f : \Sigma^{\mathbb{Z}} \rightarrow \Sigma^{\mathbb{Z}}$ be a CA that is not nilpotent but is nilpotent on periodic points. Take a graph G consisting of copies of cycle graphs C_n . Then the local rule of f induces a CA on colorings of G , and this CA is SAN.
- ▶ The previous example is uniformly asymptotically nilpotent (each individual cycle is all-zero after a bounded number of steps). There are also examples in this framework which are non-uniformly asymptotically nilpotent.

SAN CA on road-colored graphs

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

The questions

Experimentation

1D

d D

Groups and graphs

Conclusions

- ▶ Let $f : \Sigma^{\mathbb{Z}} \rightarrow \Sigma^{\mathbb{Z}}$ be a CA that is not nilpotent but is nilpotent on periodic points. Take a graph G consisting of copies of cycle graphs C_n . Then the local rule of f induces a CA on colorings of G , and this CA is SAN.
- ▶ The previous example is uniformly asymptotically nilpotent (each individual cycle is all-zero after a bounded number of steps). There are also examples in this framework which are non-uniformly asymptotically nilpotent.
- ▶ The constructions I know rely strongly on the graph being non-uniform (not a group).

SAN CA on groups

- ▶ On (full shifts on) Cayley graphs of groups, it seems difficult to construct SAN CA.

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

The questions

Experimentation

1D

d D

Groups and graphs

Conclusions

SAN CA on groups

- ▶ On (full shifts on) Cayley graphs of groups, it seems difficult to construct SAN CA.

Question

Which full shifts on finitely generated groups support cellular automata that are asymptotically nilpotent but not nilpotent? Do any?

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

The questions

Experimentation

1D

dD

Groups and graphs

Conclusions

SAN CA on groups

- ▶ On (full shifts on) Cayley graphs of groups, it seems difficult to construct SAN CA.

Question

Which full shifts on finitely generated groups support cellular automata that are asymptotically nilpotent but not nilpotent? Do any?

Conjecture

Heisenberg group is nilrigid.

SAN CA on groups

- ▶ On (full shifts on) Cayley graphs of groups, it seems difficult to construct SAN CA.

Question

Which full shifts on finitely generated groups support cellular automata that are asymptotically nilpotent but not nilpotent? Do any?

Conjecture

Heisenberg group is nilrigid.

- ▶ Seems to be solved now (work-in-progress with Ilkka Törmä).

SAN CA on groups

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

The questions

Experimentation

1D

dD

Groups and graphs

Conclusions

- ▶ On (full shifts on) Cayley graphs of groups, it seems difficult to construct SAN CA.

Question

Which full shifts on finitely generated groups support cellular automata that are asymptotically nilpotent but not nilpotent? Do any?

Conjecture

Heisenberg group is nilrigid.

- ▶ Seems to be solved now (work-in-progress with Ilkka Törmä).

Question

Is the free group nilrigid?

Revised questions

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

Question

Is there an interesting example of a nilpotent CA on a one-dimensional full shift?

The questions

Experimentation

1D

dD

Groups and graphs

Conclusions

Revised questions

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

Question

Is there an interesting example of a nilpotent CA on a one-dimensional full shift?

Question

Which one-dimensional sofic shifts are nilrigid? Is this decidable?

The questions

Experimentation

1D

dD

Groups and graphs

Conclusions

Revised questions

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

Question

Is there an interesting example of a nilpotent CA on a one-dimensional full shift?

Question

Which one-dimensional sofic shifts are nilrigid? Is this decidable?

Question

Are all two-dimensional SFTs nilrigid?

The questions

Experimentation

1D

dD

Groups and graphs

Conclusions

Revised questions

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

The questions

Experimentation

1D

dD

Groups and graphs

Conclusions

Question

Is there an interesting example of a nilpotent CA on a one-dimensional full shift?

Question

Which one-dimensional sofic shifts are nilrigid? Is this decidable?

Question

Are all two-dimensional SFTs nilrigid?

Question

Are all groups nilrigid? Is the free group nilrigid?

Other observations: variants of nilpotency

- It is known that nilpotency is not equivalent to nilpotency on periodic points, or on finite points.

Other observations: variants of nilpotency

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

The questions

Experimentation

1D

dD

Groups and graphs

Conclusions

- ▶ It is known that nilpotency is not equivalent to nilpotency on periodic points, or on finite points.
- ▶ It is known that nilpotency is not equivalent to all traces having 0s with a density 1 [Törmä, '15]

Other observations: variants of nilpotency

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

The questions

Experimentation

1D

dD

Groups and graphs

Conclusions

- ▶ It is known that nilpotency is not equivalent to nilpotency on periodic points, or on finite points.
- ▶ It is known that nilpotency is not equivalent to all traces having 0s with a density 1 [Törmä, '15]
- ▶ CA also act on measures, and 'randomization' is precisely 'asymptotic nilpotency' towards the uniform Bernoulli measure. It is open whether there are ergodic PCA where such convergence happens on all initial measures, but the convergence is non-uniform.

Other observations: variants of nilpotency

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

The questions

Experimentation

1D

dD

Groups and graphs

Conclusions

- ▶ It is known that nilpotency is not equivalent to nilpotency on periodic points, or on finite points.
- ▶ It is known that nilpotency is not equivalent to all traces having 0s with a density 1 [Törmä, '15]
- ▶ CA also act on measures, and 'randomization' is precisely 'asymptotic nilpotency' towards the uniform Bernoulli measure. It is open whether there are ergodic PCA where such convergence happens on all initial measures, but the convergence is non-uniform.
- ▶ CA on vector alphabets can be non-uniformly asymptotically nilpotent (Work in progress with Siamak Taati)

The End

Thank you for listening!

Strict asymptotic
nilpotency in
cellular automata

Ville Salo

The questions

Experimentation

1D

d D

Groups and graphs

Conclusions